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Oefeningenreeks 7C nr. 1.1

$$Df(x,y,z) = \begin{pmatrix} y^2z^3 + 2xy^3z^4 + 3x^2y^4z^5 \\ 2xyz^3 + 3x^2y^2z^4 + 4x^3y^3z^5 \\ 3xy^2z^2 + 4x^2y^3z^3 + 5x^3y^4z^4 \end{pmatrix}$$

$$D^2f(x,y,z) = \begin{pmatrix} 0 + 2y^3z^4 + 6xy^4z^5 & 2yz^3 + 6xy^2z^4 + 12x^2y^3z^5 & 3y^2z^2 + 8xy^3z^3 + 15x^2y^4z^4 \\ 2yz^3 + 6xy^2z^4 + 12x^2y^3z^5 & 2xz^3 + 6x^2yz^4 + 12x^3y^2z^5 & 6xyz^2 + 12x^2y^2z^3 + 20x^3y^3z^4 \\ 3y^2z^2 + 8xy^3z^3 + 15x^2y^4z^4 & 6xyz^2 + 12x^2y^2z^3 + 20x^3y^3z^4 & 6xy^2z + 12x^2y^3z^2 + 20x^3y^4z^3 \end{pmatrix}$$

$$D^2f(x,y,z) = \begin{pmatrix} 2y^3z^4\left(1 + 3xyz\right) & 2yz^3\left(1 + 3xyz + 6x^2y^2z^2\right) & y^2z^2\left(3 + 8xyz + 15x^2y^2z^2\right) \\ 2yz^3\left(1 + 3xyz + 6x^2y^2z^2\right) & 2xz^3\left(1 + 3xyz + 6x^2y^2z^2\right) & 2xyz^2\left(3 + 6xyz + 10x^2y^2z^2\right) \\ y^2z^2\left(3 + 8xyz + 15x^2y^2z^2\right) & 2xyz^2\left(3 + 6xyz + 10x^2y^2z^2\right) & 2xy^2z\left(3 + 6xyz + 10x^2y^2z^2\right) \end{pmatrix}$$

References